ELIMINATION OF IMAGINARIES IN FIELDS WITH A DERIVATION

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Abstract. I present some of the model theory of fields with derivation, leading to a proof of elimination of imaginaries in differentially closed fields. This property allows us to realize quotients of definable sets by definable equivalence relations using definable functions, which in this context are piecewise differential-rational functions. Along the road to this we will see quantifier elimination, a correspondence between types and ideals, some of the architecture of totally transcendental theories including Morley rank, and canonical bases. These features also appear in a simpler and more familiar form in the model theory of algebraically closed fields, so I will explain this setting in parallel with the differential setting.

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1. Introduction

The program of algebraic study of differential equations we follow here starts with the work of Liouville, Picard, and Vessiot in the 19th century on the expressibility of indefinite integrals and solutions to linear differential equations in terms of elementary functions. Differential algebra as such was developed using the methods of commutative algebra and algebraic geometry by Ritt and Kolchin. Many of the results of this work can be found in [Kol73].

Starting with the work of Robinson and Blum, fields with derivation were studied from the point of view of model theory. The theory of differentially closed fields of characteristic zero was isolated as an analogue for fields with derivation to the theory of algebraically closed fields in [Rob59]. Specifically, this theory was shown to be the model completion of the theory of differential rings of characteristic zero. It was also shown in [Blu69] to be totally transcendental in the sense of [Mor65].

In [She78] Shelah introduced *imaginaries*, equivalence classes of elements which can serve as canonical codes for definable sets. Poizat elaborated on this in [Poi83], showing that in some familiar classes of structures, these imaginary canonical codes are always interdefinable with tuples of ordinary elements. This phenomenon, known as *elimination of imaginaries*, allows a quotient by a definable equivalence relation to be realized as definable function. An instance of this in the theory of algebraically closed fields is the realization of the quotient $S_n \setminus \mathbb{A}^n$ using symmetric polynomials.

Poizat proved in [Poi83] that the theory of differentially closed fields of characteristic 0, DCF_0, admits elimination of imaginaries in this sense. In this paper we will develop the model theory of differentially closed fields towards a proof of this theorem. In outline the proof here follows Poizat's original proof, as well as the exposition in [MMP96].

In Section 2 we introduce differential rings, in particular rings of differential polynomials, and study their algebra. In Section 3 we use these algebraic results to prove quantifier elimination and establish some of its consequences, including a classification of types in a single variable. In Section 4 we introduce two tools that give us leverage over definable sets and types in multiple variables. The first of these is Morley's notion of a totally transcendental theory and the associated ordinal rank on formulas and types. The second is an algebraic description of types in terms of prime differential ideals in rings of differential polynomials, and a primary decomposition theorem concerning these ideals. In Section 5 we establish general model theoretic criteria for elimination of imaginaries in terms of canonical bases, and then apply these to prove that the theory DCF_0 has this property. Finally, in Sections 6 and 7 we see some examples of elimination of imaginaries, and go over some open questions regarding differentially closed fields.

2. Rings with Derivation

2.1. **Differential Rings and Fields.**

A derivation on a ring R is an additive homomorphism $\partial : R \to R$ that satisfies the Leibniz rule:

$$\partial(x \cdot y) = \partial x \cdot y + x \cdot \partial y$$

Familiar formulas for derivatives of quantities which are polynomial, rational, or (separably) algebraic in known quantities can be derived from this. In this paper, we will consider differential rings which are integral domains and contain $\mathbb{Q}$. Such differential rings include:

(1) A field k of characteristic 0 equipped with the zero derivation.
(2) A polynomial ring $k[t]$, equipped with the unique derivation ∂ satisfying $\partial|_k = 0$ and $\partial(t) = 1$.
(3) More generally, given any differential ring R, and any element $f \in R[t]$, the ring $R[t]$ equipped with the unique derivation extending the derivation on R and sending $\partial : t \mapsto f$.
(4) The field of rational functions $k(t)$, i.e. the field of fractions of the polynomial ring in (2).
(5) The ring of formal power series $k[[t]]$, or Laurent series $k((t))$, or Puiseux series $k\langle\langle t\rangle\rangle$.
(6) The ring of holomorphic functions $\mathcal{O}(U)$ on a connected open subset $U \subset \mathbb{C}$.

In this section, we will introduce some of the basic elements of the algebra of these rings. For a differential ring R, the set of *constants* of R is the kernel of ∂. We denote this by C_R.

Lemma 1. *C_R is a subring of R, which is a subfield if R is a field, and is integrally closed in R.*

Proof Sketch. That C_R is a subring follows from the product rule, and that it is a subfield follows from the quotient rule. Let $r \in R$ be integral over C_r with equation of integral dependence $p(r) = 0$, assumed to be of minimal degree. Letting $q(t)$ be obtained by formally differentiating $p(t)$ in t, we have

$$0 = (\partial r) \cdot q(r)$$

$\frac{1}{n}q(t)$ is a monic polynomial over C_R of less than $p(t)$, so $q(r) \neq 0$. Therefore, $\partial r = 0$, so $r \in C_R$. $\square$

For R a differential ring, an ideal $I \subset R$ is called a *differential ideal* if $\partial I \subset I$. This implies that the derivation descends to a derivation on R/I. Given elements $f_1, \dots, f_r \in R$, we let

$$\langle f_1, \dots, f_r \rangle = (f_i^{(j)})_{i \leq r, j \in \mathbb{N}}$$

This is the smallest differential ideal containing $f_1, \dots, f_r$.

To describe differential algebraic relations between elements of differential rings, we will use a construction analogous to the use of ideals in polynomial rings to describe algebraic relations.

Definition. Let R be a ring with derivation ∂. The ring $R\{X\}$ of differential polynomials R is the polynomial algebra

$$R[X^{(0)}, X^{(1)}, X^{(2)}, \dots]$$

We extend the derivation ∂ to $R\{X\}$ by setting $\partial X^{(i)} = X^{(i+1)}$. We call the elements of $R\{X\}$ *differential polynomials over* R.

The *order* of a differential polynomial f is the largest n such that $X^{(n)}$ occurs in f. By convention constant polynomials (elements of R) have order -1. We view a polynomial $f \in R\{X\}$ of order n as a polynomial in $X^{(n)}$ with coefficients in $R[X^{(0)}, \dots, X^{(n-1)}]$. By the *degree* of f, we mean its degree as a polynomial in $X^{(n)}$ in this sense. Given two differential polynomials f and g, we write

$$f \ll g$$

if either g has greater order than f, or they have the same order and g has greater degree. This is a well founded ordering on differential polynomials, so we can argue by induction on it.

Continuing to view f as a polynomial in $X^{(n)}$ with coefficients in $R[X^{(0)}, \dots, X^{(n-1)}]$, we define the *initial* of f to be its leading coefficient in $X^{(n)}$,

$$i_f \in R[X^{(0)}, \dots, X^{(n)}]$$

This is to say, f can be written

$$f = i_f \cdot (X^{(n)})^d + g$$

where $g \ll f$. We define the *separant* to be the formal derivative

$$s_f = \frac{\partial f}{\partial X^{(n)}} \in R[X^{(0)}, \dots, X^{(n)}]$$

This formal derivative does not depend on the derivation on $R\{X\}$, it is obtained by applying the usual formula to differentiate a polynomial, treating its coefficients as constant. One motivation for the separant is algebraic, as it allows us to describe the leading terms of the derivatives of f in the differential ring $R\{X\}$.

Lemma 2 (Leading term of derivative)**.** *For $f \in R\{X\}$ of order n, for any $j > 0$ we can write*

$$\partial^j f = s_f \cdot X^{(n+j)} + g$$

where g has lower order than $n + j$.

Proof. Write

$$f = \sum_i a_i(X^{(0)}, \dots, X^{(n-1)})(X^{(n)})^i$$

Applying the product rule to differentiate this, we have

$$\partial f = \sum_i \left[a_i \cdot i(X^{(n)})^{i-1} X^{(n+1)} + [\partial a_i]\,(X^{(n)})^i \right] = s_f \cdot X^{(n+1)} + g$$

where g is of order at most n. For $j > 1$, we obtain the formula directly by induction on j. $\square$

Another motivation for introducing these quantities is analytic. The following theorem can motivate many of the algebraic definitions in this paper.

Theorem 1 (Cauchy-Kovalevskaya, this particular form follows from the implicit function theorem and Section 1.3 of [Ste17])**.** *Let $U \subset \mathbb{C}$ be an open set, $\mathcal{O} = \mathcal{O}(U)$ the ring of holomorphic functions on U, $f(x_0, \ldots, x_n) \in \mathcal{O}[x_0, \ldots, x_n]$ an irreducible polynomial with coefficients in $\mathcal{O}$. Consider the differential equation*

$$(\star) \qquad f(y^{(0)}, y^{(1)}, \ldots, y^{(n)}) = 0$$

Define

$$s_f = \frac{\partial f}{\partial x_n} \in \mathcal{O}[x_0, \ldots, x_n]$$

and let

$$i_f(y_0, \ldots, y_{n-1}) \in \mathcal{O}[x_0, \ldots, x_{n-1}]$$

be the initial coefficient of f as a polynomial in x_n. Suppose $z_0 \in U$, and $w_0, \ldots, w_n \in \mathbb{C}$ satisfy

$$f(w_0, \ldots, w_n)(z_0) = 0$$

$$i_f(w_0, \ldots, w_{n-1})(z_0) \neq 0$$

$$s_f(w_0, \ldots, w_n)(z_0) \neq 0$$

Then there is a smaller open set $z_0 \in V \subset U$ and a $y \in \mathcal{O}(V)$ satisfying $(\star)$ with

$$y^{(i)}(z_0) = w_i$$

for $i \in \{0, \ldots, n\}$. Moreover, any two such solutions agree on their common domain.

We conclude with some definitions regarding the vanishing sets of differential polynomials. Let k be a differential field. We say that a subset of k^n is *Kolchin closed* if it has the form

$$V_\partial(S) = \{x \in k^n \mid f(\overline{x}) = 0 \text{ for each } f \in S\}$$

for some subset $S \subset k\{X_1, \ldots, X_n\}$. The collection of these sets form the closed sets of a topology on k^n, called the Kolchin topology. This is analogous to the Zariski topology on k^n when k is an ordinary field. We will see in Section 4 that any Kolchin closed set can in fact be defined by finitely many differential polynomials, and so is a definable set in model theoretic terms.

We say that a set is *Kolchin constructible* if it is a boolean combination of Kolchin closed sets. These sets are exactly those defined by quantifier free formulas in the language of differential rings. Later on, when we are working in differentially closed fields, we will have a quantifier elimination result which implies that any definable set is Kolchin constructible.

2.2. **Ritt-Raudenbush Theory in One Variable.** Ritt-Raudenbush theory gives a description of prime differential ideals in rings of differential polynomials. The logical motivation for this theory in our context is that prime differential ideals are essentially the same thing as quantifier free types in differential fields. For the model theory of differentially closed fields, we will need descriptions of these quantifier free types.

The main theorem is the following.

Theorem 2 (Characterization of Prime Differential Ideals)**.** *Let k be a differential field.*

(1) *If $f \in k\{X\}$ is an irreducible polynomial, then the set*

$$I(f) = \{g \in k\{X\} \mid s_f^n g \in \langle f \rangle \text{ for some } n\}$$

is a prime differential ideal.

(2) *Every prime differential ideal in $k\{X\}$ has the form $I(f)$ form some f, which can be chosen to be a $\ll$-minimal nonzero element of the ideal.*

(3) *$I(f)$ is the unique prime differential ideal containing f and no nonzero differential polynomials of lower order.*

This description of prime differential ideals is necessary because simply taking $\langle f \rangle$ for irreducible f does not always work. Letting $f = (X^{(1)})^2 - 2X^{(0)}$, we have

$$\langle f \rangle \ni \partial f = 2X^{(1)}(X^{(2)} - 1)$$

We will show that neither factor is in this ideal, so it is not prime. This is a purely ring theoretic question, so we can forget about the derivation. If we plug in $X^{(j)} = 0$ for $j > 2$, we have

$$\partial^2(f) = 2X^{(2)}(X^{(2)} - 1)$$

$$\partial^j(f) = 0 \;:\; j > 2$$

Thus letting $x = X^{(0)}$, $y = X^{(1)}$ and $z = X^{(2)}$ it suffices to show that

$$2y \notin (y^2 - 2x, 2y(z-1), 2z(z-1)) \subset k[x, y, z]$$

$$(z-1) \notin (y^2 - 2x, 2y(z-1), 2z(z-1)) \subset k[x, y, z]$$

We can plug in $x = y = z = 0$ to see that the ideal does not contain $z - 1$. Similarly, we can plug in $x = \frac{1}{2}$, $y = z = 1$ to see that the ideal does not contain $2y$.

We now set about proving Theorem 2.

Lemma 3 (Differential Division with Remainder)**.** *Let $f \in R\{X\}$ satisfy* $\operatorname{ord}(f) \geq 0$*, and $g \in R\{X\}$. There are are integers m, k and $q \ll f$ such that*

$$i_f^k \cdot s_f^m \cdot g \equiv q \mod \langle f \rangle$$

Proof Sketch. The proof proceeds in two steps. Let n be the order of f.

(1) There is a p of order $\leq n$ such that

$$s_f^m \cdot g \equiv p \mod \langle f \rangle$$

(2) There is a $q \ll f$ such that

$$i_f^k \cdot p \equiv q \mod \langle f \rangle$$

Step (2) is an application of ordinary polynomial division with remainder in the polynomial ring

$$A[X^{(n)}] = R[X^{(0)}, \ldots, X^{(n-1)}][X^{(n)}]$$

For step (1), we argue by induction on g with respect to $\ll$ order, assuming $\operatorname{ord}(g) = n + j > n$. The point is that using Lemma 2 we can kill the leading term of $s_f \cdot g$ by subtracting a multiple of $\partial^j f$. ☐

Lemma 4. *For any R and $f \in R\{X\}$, $I(f)$ is a differential ideal.*

Proof. $I(f)$ is an ideal because it is the preimage in $R\{X\}$ of the ideal generated by $\{f^{(i)} \mid i < \omega\}$ in the ring $R\{X\}[s_f^{-1}]$. To show that it is a differential ideal, let g be such that $s_f^n \cdot g \in \langle f \rangle$. Then

$$\partial \left[s_f^{n+1} \cdot g \right] = (n+1) \cdot [\partial s_f]\, s_f^n \cdot g + s_f^{n+1} \cdot [\partial g] \in \langle f \rangle$$

and so $s_f^n \cdot [\partial g] \in \langle f \rangle$, $\partial g \in I(f)$. ☐

Lemma 5. *If f is irreducible,* $\operatorname{ord}(f) \geq 0$*, and $g \in I(f)$ satisfies* $\operatorname{ord}(g) \leq \operatorname{ord}(f)$*, then f divides g.*

Proof. Let $n = \operatorname{ord}(f)$. If $g \in I(f)$, then we can write

$$s_f^r \cdot g = \sum_{j \leq N} h_j \cdot \partial^j f$$

We first show that $s_f^{r'} \cdot g \in (f)$ for some r'. For $N > 0$, $X^{(n+N)}$ does not occur on the left, so we can plug in arbitrary values for it on the right. Applying Lemma 2 we write

$$\partial^N f = s_f \cdot X^{(n+N)} + g$$

where $X^{(n+N)}$ does not occur in g, then substitute $-g/s_f$ for $X^{(n+N)}$, which makes the last term in the sum 0. Clearing denominators, we have

$$s_f^{r'} \cdot g = \sum_{j \leq N-1} h'_j \cdot \partial^j f$$

Appealing to induction on N, this implies that for some r', $s_f^{r'} \cdot g \in (f)$. Now as f is irreducible and f does not divide s_f, we conclude that f divides g. ☐

Proof of Theorem 2. Let k be a differential field, and $f \in k\{X\}$ irreducible of order $n \geq 0$.

(1) It remains to show that $I(f)$ is prime. Suppose $g \cdot h \in I(f)$. Applying step (1) of Lemma 3, there are p and q of order $\leq n$ such that $s_f^k g \equiv p \mod \langle f \rangle$ and $s_f^m \equiv q \mod \langle f \rangle$. We then have that $p \cdot q \in I(f)$, so as $p \cdot q$ has order $\leq n$, f divides $p \cdot q$. Now as f is irreducible, it divides either p and q. This implies that either f or g is in $I(f)$, as required.

(2) Let $\mathfrak{p}$ be a prime differential ideal in $k\{X\}$, and let f be a $\ll$-least element of $\mathfrak{p}$. It follows that $s_f \notin \mathfrak{p}$, so $I(f) \subset \mathfrak{p}$ as $\mathfrak{p}$ is prime. On the other hand, if $g \in \mathfrak{p}$, then by Lemma 3 for some m, k and $p \in k\{X\}$ of order at most n

$$i_f^k \cdot s_f^m \cdot g \equiv q \mod \langle f \rangle$$

Because i_f and s_f are not in $I(f)$ and $I(f)$ is prime, to show that $g \in I(f)$ it suffices to show that $q \in I(f)$. We have $q \in \mathfrak{p}$, but as $q \ll f$ and f is $\ll$-minimal among nonzero elements of $\mathfrak{p}$, we must have $q = 0$.

(3) Let $\mathfrak{p}$ be a prime differential ideal containing f and no nonzero differential polynomials of lower order. Then $\mathfrak{p} = I(g)$ for some irreducible differential polynomial g, where g is $\ll$-minimal among nonzero elements of $\mathfrak{p}$. This implies that f and g have the same order, so by Lemma 5, g divides f. As f is irreducible, this implies that f is a unit multiple of g, so f is also $\ll$-minimal in $\mathfrak{p}$. Therefore, $\mathfrak{p} = I(f)$ as shown in (2).

□

3. The Theory of Differentially Closed Fields

In this section we introduce the first order theory of differentially closed fields, a class of differential rings containing solutions to all consistent systems of algebraic differential equations which will be our main object of study.

3.1. **The Axioms.** DCF_0 is a first order theory in the language of differential rings, with axioms asserting the existence of sufficiently many solutions to all algebraic differential equations. The meaning of "sufficiently many solutions" that we use will be that the solution set to an order n equation is large enough that it is not contained in the solution set of an order m equation for $m < n$. This can be understood as a reflection of the fact that initial conditions for a local analytic solution to a lower order equation have fewer independent paramters.

The axioms of DCF_0 consist of the following statements in the first order language of differential rings:

(1) Axioms stating that the structure is a field of characteristic 0.
(2) Axioms stating that the symbol ∂ is a derivation on this field.
(3) Axioms stating that the structure is algebraically closed as a field. For each n, we have the sentence

$$(\forall y_0 \dots y_{n-1})(\exists x)(x^n + y_{n-1}x^{n-1} + \cdots + y_0 = 0)$$

(4) For each pair of differential polynomials

$$f, g \in \mathbb{Z}[y_0, \dots, y_N]\{X\}$$

such that $\mathrm{ord}_X(f) > \mathrm{ord}_X(g)$, the sentence

$$(\forall y_0, \dots, y_N)(\text{the leading coefficients of } f \text{ and } g \text{ do not vanish} \to (\exists x)(f(x) = 0 \wedge g(x) \neq 0))$$

The theory ACF_0 of algebraically closed fields of characteristic 0 consists of the axioms in (1) and (3), in the language of rings. The axioms in (4) imply that for $k \models \mathsf{DCF}_0$, given any

$$f, g \in k\{X\}$$

such that $\mathrm{ord}(f) > \mathrm{ord}(g)$ and $g \neq 0$, there is an $\alpha \in k$ such that

$$f(\alpha) = 0 \wedge g(\alpha) \neq 0$$

In other words, the solution set of f is not contained in that of g. Taking $g = 1$, this ensures in particular that all non-constant differential polynomials have zeros in k. Because the field of constants $C \subset k$ is integrally closed in k, and k is algebraically closed as a field, C is algebraically closed as a field.

Finding explicit models of DCF_0 is harder than finding explicit models of ACF_0. We will say more about this and the relationship between elements of models of DCF_0 and analytic solutions to differential equations in Section 3.4. At this point, we can at least check that DCF_0 is a consistent theory which is compatible with the structure of any differential ring under consideration.

Theorem 3 (Model Existence for DCF_0)**.** *Let (R, ∂) be a differential ring which is an integral domain of characteristic 0. Then there is an embedding $R \hookrightarrow K$ into a model K of DCF_0.*

Proof. We may first replace R with its field of fractions k, so assume that $R = k$ is a differential field. We construct a field $k^{(1)}$ as follows. Enumerate all nonzero differential polynomials $f \in k\{X\}$ as $(f_\alpha)_{\alpha<\beta}$. We then construct k_α be recursion, starting with $k_0 = k$. Given k_α, we construct $k_{\alpha+1}$ as follows. Take an irreducible factor h_α of f_α in $k_\alpha\{X\}$ of order equal to f_α, and consider the prime differential ideal $I(h_\alpha)$. Then

$$R_{\alpha+1} = k_\alpha\{X\}/I(h_\alpha)$$

is a domain containing an element x satisfying f_α and not satisfying any lower order differential polynomial over k_α. Let $k_{\alpha+1}$ be the field of fractions of $R_{\alpha+1}$. At limit stages we take unions, and we set $k^{(1)} = \cup_{\alpha<\beta} k_\alpha$. Now start from $k^{(1)}$ and construct likewise $k^{(2)}, k^{(3)}, \ldots$. Finally, let $k^{(\omega)} = \cup_{n<\omega} k^{(n)}$. $k^{(\omega)}$ is then a model of DCF_0. □

One can say more, there is a model of DCF_0 which is related to any differential field of characteristic 0 in a way analogous to the relationship between a field and its algebraic closure, which we call its differential closure. We will return to this in Section 4.

3.2. **Quantifier Elimination and Types in One Variable.** We now begin our study of definable sets and types in DCF_0. The first step of this is the following theorem.

Theorem 4 (Quantifier Elimination in DCF_0)**.** *DCF_0 admits quantifier elimination in the language of differential rings. Any formula is equivalent over DCF_0 to a quantifier free formula. In particular, any definable set (with parameters) is Kolchin constructible.*

To prove this, we will use the following general model theoretic criterion for quantifier elimination.

Proposition (Criterion for quantifier elimination)**.** *Let T be a theory in a language L. The following are equivalent:*

(1) *T admits quantifier elimination in the language L.*
(2) *For any A, B models of T where B is ω-saturated, sharing a common finitely generated L-substructure $C = \langle \overline{c} \rangle$ like so:*

$$\begin{tikzcd} A & & B \\ & C \arrow[ul, dash] \arrow[ur, dash] & \end{tikzcd}$$

if $\varphi(x, \overline{y})$ is quantifier free we have the implication

$$A \models (\exists x)\varphi(x, \overline{c}) \Longrightarrow B \models (\exists x)\varphi(x, \overline{c})$$

The meaning of this criterion is that quantifier free information common to A and B determines whether or not an existential statement is satisfied. This allows one to prove inductively that the satisfaction of any formula is determined by quantifier free information. A proof is given in Appendix A. To apply this criterion to DCF_0 we will use our understanding of prime differential ideals in rings $k\{X\}$.

Proof of quantifier elimination in DCF_0. Let K and L be models of DCF_0 with L ω-saturated, and $R = \mathbb{Z}\langle \overline{c} \rangle$ be a finitely generated sub-differential ring of both K and L. Suppose that $K \models (\exists x)\varphi(x, \overline{c})$ for some quantifier free formula φ, witnessed by $\alpha \in K$. The field of fractions k of R is contained in K and L because they are fields. To show that $L \models (\exists x)\varphi(x, \overline{c})$ we will find an embedding

$$k\langle \alpha \rangle \hookrightarrow L$$

over k. It suffices to find a $\beta \in L$ such that $\alpha \mapsto \beta$ extends to such an embedding. Consider the prime differential ideal

$$\mathfrak{p} = I(\alpha/k) = \{f(X) \in k\{X\} \mid f(\alpha) = 0\}$$

The required property of β is that $I(\beta/k) = \mathfrak{p}$. By Theorem 2, there is some irreducible polynomial $f \in k\{X\}$ such that $\mathfrak{p} = I(f)$ is the unique prime differential ideal containing f and no differential polynomials of lower order. This implies that $I(\beta/k) = \mathfrak{p}$ provided that $f(\beta) = 0$ and for every $g \in k\{X\}$ of lower order, $g(\beta) \neq 0$. That is, β needs to satisfy the set of quantifier free formulas

$$\{f(x) = 0\} \cup \{g(x) \neq 0 \mid \mathrm{ord}(g) < \mathrm{ord}(f)\}$$

This is a partial type over a finite set of parameters $\bar{c}$, as k is generated by $\bar{c}$ as a differential field. Because L is ω-saturated, to show that this is realized in L it suffices to show that every finite subset of it is realized in L. Given a finite subset involving $g_1, \ldots, g_r$, we can find a γ in L such that

$$f(\gamma) = 0$$

$$\prod_{i \leq r} g_i(\gamma) \neq 0$$

because $L \models \mathsf{DCF}_0$. This completes the proof. □

Remark. A similar but simpler argument proves quantifier elimination in ACF_p for any p, using the fact that $k[t]$ is a principal ideal domain for k a field.

A first corollary of quantifier elimination is a description of all types in one variable over certain parameter sets.

Corollary (1-Types in DCF_0)**.** *Let k be a differential field contained in a model of DCF_0. Let $S_1(k)$ be the set of 1-types with parameters in k, and $\mathrm{Spec}_\partial(k\{X\})$ be the set of prime differential ideals in $k\{X\}$. Then the map*

$$p \longmapsto \{f \in k\{X\} \mid p(x) \vdash f(x) = 0\} = I(p)$$

is a bijection

$$S_1(k) \longleftrightarrow \mathrm{Spec}_\partial(k\{X\})$$

In particular, any element p of $S_1(k)$ has one of the following forms

1. *For some irreducible $f \in k\{X\}$, p is axiomatized by the formulas $f(x) = 0$ and $g(x) \neq 0$ for all g of lower order than f. We call such types differentially algebraic, and f a minimal polynomial for the type.*
2. *p is axiomatized by $g(x) \neq 0$ for all nonzero $g \in k\{X\}$. We call this the type of a differentially transcendental element.*

Proof. $I(p)$ can be obtained by letting α be a realization of p in some larger differential field, and taking $I(p) = I(\alpha/k)$. This description shows that $I(p)$ is a prime differential ideal. Now we show that $p \mapsto I(p)$ is surjective. For $\mathfrak{p} \in \mathrm{Spec}_\partial(k\{X\})$ define

$$R = k\{X\}/\mathfrak{p}$$

By the model existence theorem, there is a $K \models \mathsf{DCF}_0$ with $R \subset K$. The element $\alpha \in K$ coming from $X^{(0)} \in k\{X\}$ realizes a type $p \in S_1(k)$ with $I(p) = \mathfrak{p}$. (A somewhat subtle point here: let L be the model of DCF_0 from which we got k. Types in $S_1(k)$ need to be consistent with all the properties elements of k have *in* L. Our construction works because L and K agree on all these properties by quantifier elimination.) We now show that $p \mapsto I(p)$ is injective. If $I(p) = I(q)$ then p and q have the same restriction to quantifier free formulas, so are equal by quantifier elimination. The description of types then follows from our earlier description of prime differential ideals in $k\{X\}$. □

Remark. A parallel argument shows that in ACF_p, given a field k contained in a model of ACF_p, there is a bijection

$$S_1(k) \longleftrightarrow \mathrm{Spec}(k[t])$$

Another corollary of quantifier elimination is that DCF_0 is a complete theory.

Corollary. DCF_0 *is a complete theory in the language of differential rings.*

Proof. For any sentence φ, $\mathsf{DCF}_0 \vdash \varphi \leftrightarrow \theta$, where θ is quantifier free. For any differential rings $R \subset S$, R and S agree about the truth of θ. Because any model of DCF_0 contains $\mathbb{Z}$ with the 0 derivation as a differential subring, any two models of DCF_0 agree on the truth of θ, and so agree on the truth of φ. □

Another immediate consequence is model completeness. See Appendix A for a general model theoretic introduction to this notion.

Theorem 5. DCF_0 *is model complete. If $K \subset L$ are models of DCF_0, then $K \preceq L$, K is an elementary substructure of L.*

Together with model existence, this implies in particular that any system of equations and inequalities satisfied in a differential field extending $K \models \mathsf{DCF}_0$ is already satisfied in K.

3.3. **Definable Functions.** As another application of quantifier elimination, we will describe functions whose graphs are definable in models of DCF_0. This will be important for the interpretation of our results on elimination of imaginaries.

Theorem 6 (Characterization of Definable Functions)**.** *Let $K \models \mathsf{DCF}_0$, X and Y be definable sets over K, and $f : X \to Y$ be a definable function. Then there is a decomposition of X*

$$X = X_1 \sqcup \cdots \sqcup X_r$$

into Kolchin constructible subsets, such that

$$f_i = f \restriction X_i$$

is a differential rational function, its coordinates are ratios of differential polynomials in the coordinates of the input.

We will deduce this "global" characterization of definable functions using compactness and a "local" characterization of definable elements. Given a subset A of a model K of DCF_0, we define $\mathrm{dcl}(A)$ to be the set of elements of K definable using parameters from A, and $\mathrm{acl}(A)$ to be set of elements b such that there is a formula $\varphi(x)$ with parameters from A such that $K \models \varphi(b)$ and the solution set of $\varphi(x)$ is finite.

Lemma 6 (Characterization of acl and dcl)**.** *Let k be the differential field generated by A. Then*

(1) $\mathrm{dcl}(A) = k$
(2) $\mathrm{acl}(A) = \overline{k}$*, the field theoretic algebraic closure of k.*

Proof. We prove (2) and then deduce (1) as a consequence. For (2), it is clear that $\overline{k} \subset \mathrm{acl}(A)$, as the polynomial witnessing algebraicity over k can be turned into a formula with parameters in A. For the other direction, suppose $b \in \mathrm{acl}(A)$ and consider the prime differential ideal $I(f) \subset k\{X\}$ corresponding to $\mathrm{tp}(b/k)$. If f has order > 0, let K be a differentially closed field containing k, and let g be an irreducible factor of f in $K\{X\}$ of order > 0. Let β realize the type in $S_1(K)$ corresponding to $I(g)$. Then β is a zero of f and no elements of $k\{X\}$ of lower order, so $I(\beta/k) = I(f) = I(b/k)$. Therefore, $\mathrm{tp}(\beta/k) = \mathrm{tp}(b/k)$, but $\beta \notin K$. This is a contradiction, because $\mathrm{tp}(\beta/k)$ has finitely many realizations, all of which must be in K.

Now we deduce (1). Again it is clear that $k \subset \mathrm{dcl}(A)$. Conversely, if $b \in \mathrm{dcl}(A)$, $b \in \overline{k}$ by (2). If $\beta \notin k$, then consider the minimal polynomial of b over k. Let $b' \neq b$ be a root of this polynomial in $\overline{k}$. Then $\mathrm{tp}(b'/k) = \mathrm{tp}(b/k)$ (look at the corresponding ideals), which contradicts the fact that b is definable. □

Proof of Theorem 6. Let $f : X \to Y$ be a definable function, defined over a set of parameters B. Looking at each coordinate of f separately, it suffices to consider the case of $f : X \to K$, where K is our model of DCF_0. Consider the set of formulas

$$\Gamma(\overline{x}) = \{\overline{x} \in X\} \cup \{f(\overline{x}) \neq g(\overline{x}) \mid g \text{ a differential rational function with coefficients from } B\}$$

If this is consistent with $E(K)$, there is a differentially closed field L extending K and an $\overline{a} \in X(L)$ such that $f(\overline{a}) \neq g(\overline{a})$ for any differential rational function g with coefficients in B. This implies that $f(\overline{a})$ is not in the differential field generated by B and $\overline{a}$, which is contradicts the lemma. Because $\Gamma(\overline{x})$ is inconsistent with $E(K)$, some finite subset of it is. This means that there are some finitely many differential rational functions $g_1, \ldots, g_r$ such that on any input, f agrees with one of the g_i. We can then choose definable (so Kolchin constructible) loci X_i such that f agrees with g_i on X_i. □

Remark. Definable functions in ACF_0 can be shown to be piecewise rational. In positive characteristic p, we need to allow pth roots as well.

3.4. **Models.** In this section, we will say a bit about models of DCF_0 and their relationship with analytic solutions to differential equations. Given a connected open subset U of the complex plane, let $\mathcal{M}(U)$ be the field of mereomorphic functions on U. The following theorem can be viewed as an extension of the Cauchy-Kovalevskaya theorem stated earlier.

Theorem 7 (Seidenberg Embedding Theorem [Sei58])**.** *Suppose that k is a countable subfield of $\mathcal{M}(U)$, and K/k is a finitely generated extension of differential fields. Then there is a connected open $V \subset U$ and an embedding $\iota : K \hookrightarrow \mathcal{M}(V)$ over k, in the sense that the following diagram commutes*

$$\begin{tikzcd} k \arrow[r] \arrow[d] & \mathcal{M}(U) \arrow[d, "\varrho"] \\ K \arrow[r, "\iota"] & \mathcal{M}(V) \end{tikzcd}$$

where ϱ is the restriction map. Moreover we may take V to be a disk whose closure is contained in U.

Let k be a countable model of DCF_0 with field of constants C. Starting with an embedding

$$C \hookrightarrow \mathbb{C} \subset \mathcal{M}(\mathbb{C})$$

we can repeatedly apply the theorem to get a descending sequence of disks

$$V_1 \supset V_2 \supset V_3 \supset \dots$$

each containing the closure of the next and embeddings

$$\iota_i : k_i \hookrightarrow \mathcal{M}(V_i)$$

where $C \subset k_1 \subset k_2 \subset \dots$ is an increasing sequence of subfields of k such that $\cup_i k_i = k$ and the diagrams

$$\begin{tikzcd} k_i \arrow[r, "\iota_i"] \arrow[d] & \mathcal{M}(V_i) \arrow[d, "\varrho"] \\ k_{i+1} \arrow[r, "\iota_{i+1}"] & \mathcal{M}(V_{i+1}) \end{tikzcd}$$

all commute. The condition on the V_i imply that their intersection is nonempty. Take z_0 to be a point of the intersection. The ι_i glue together to give an embedding

$$k \hookrightarrow \mathcal{M}_{z_0}$$

where $\mathcal{M}_{z_0}$ is the field of germs of mereomorphic functions at z_0.

Unfortunately, while it has differentially closed subfields, $\mathcal{M}_{z_0}$ is not itself differentially closed. To see this, we may take $z_0 = 0$. Consider the germ of the function z^{-1} at 0. If $\mathcal{M}_0$ was to be differentially closed, it would contain an f such that $\partial f = z^{-1}$. This is impossible, as z^{-1} has no primitive on any punctured neighborhood of 0.

4. Types, Rank, and Stability

In this section we will build on our understanding of 1-types in DCF_0 towards an understanding of types in general. We will show that types are controlled in a strong sense by single formulas, and that they can be assigned dimension-like ranks. We will then study the algebraic counterparts of types in multiple variables, differential ideals in rings of multi-variable differential polynomials. This will give us the tools necessary to prove elimination of imaginaries in the next section.

4.1. **Totally Transcendental Theories.** In his thesis [Mor65], Michael Morley proved that whenever a countable theory has a unique model of some uncountable cardinality, there are good structural reasons for this in the theory, and these imply in particular that it has a unique model of any uncountable cardinality. In doing so, he defined a class of theories whose types admit simple descriptions. It turns out that to control the complexity of types, it suffices to control how many of them there can be.

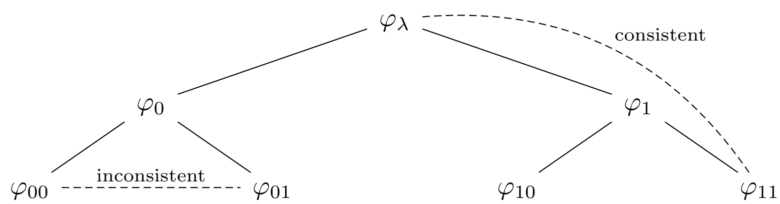
Theorem 8 (Characterization of Totally Transcendental Theories)**.** *Let T be a theory in a countable language. The following are equivalent.*

(1) *For any countable set of parameters A in any model of T, the set of 1-types $S_1(A)$ is countable.*

(2) *For any countable set of parameters A and any n, $S_n(A)$ is countable.*
(3) *There does not exist a family of formulas with parameters $\{\varphi_\eta(\overline{x})\}$ indexed by finite binary strings $\eta \in 2^{<\omega}$, such that*
   (a) *Each φ_η is consistent.*
   (b) *If $\eta \prec \sigma$ then $\varphi_\sigma \vdash \varphi_\eta$*
   (c) *For each η, $\varphi_{\eta\frown 0} \wedge \varphi_{\eta\frown 1}$ is inconsistent.*

We call any countable theory with these properties totally transcendental.

We can visualize the configuration in (3) as an infinite binary tree, whose first 3 levels are depicted below:



We will see during this section and the next how the absence of this tree configuration constrains the complexity of types. (1) and (2) are generally taken as the definition of *ω-stability*. This is a strong version of the more general property of stability, which likewise has many consequences for the possible complexity of types.

Proof Sketch. We will prove (1) implies (2) implies (3) implies (1). Assuming (1), (2) follows by by induction on n, by considering the map

$$\varrho : S_{n+1}(A) \to S_n(A)$$
$$p(x_0, \ldots, x_n) \mapsto \{\varphi(x_1, \ldots, x_n) \mid p \vdash \varphi\}$$

For a type $q \in S_n(A)$ realized by $(a_1, \ldots, a_n)$, there is a one to one correspondence between types $p(x_0, \ldots, x_n) \in \varrho^{-1}q$ and 1-types in $r(x_0) \in S_1(A \cup \{a_1, \ldots, a_n\})$, of which there are countably many by (1).

To prove that (2) implies (3), we assume that (3) fails and prove that (2) fails. Let $\varphi_\eta(\overline{x}) = \varphi_\eta(\overline{x}, \overline{a}_\eta)$ witness that (3) fails. Let A be the set of all parameters appearing in $\overline{a}_\eta$ for some η. The infinite branches of the tree yield continuum many inconsistent partial types over A, extending to continuum many elements of $S(A)$.

Finally, we prove that (3) implies (1), by showing that if (1) fails, (3) fails. Suppose A is a countable parameter set witnessing the failure of (1). We call a formula $\varphi(x)$ with parameters in A "large" if there are uncountably many types in $S_1(A)$ containing φ. By assumption, the formula $x = x$ is large. I claim that if φ is large, there is a formula $\psi(x) \in L(A)$ such that both $\varphi \wedge \psi$ and $\varphi \wedge \neg\psi$ are large. If not, then for any ψ, exactly one of these is large. The types in $S(A)$ containing φ are of two kinds. There are countably many which contain a small formula $\varphi \wedge \psi$ and one which contains exactly those ψ such that $\varphi \wedge \psi$ is large. This gives countably many in total, a contradiction. We conclude that for any large φ, there is a ψ such that $\varphi \wedge \psi$ and $\varphi \wedge \neg\psi$ are both large. This allows us to build the configuration witnessing the failure of (3) by recursion. ☐

Theorem 9. DCF_0 *is totally transcendental.*

Proof. Given a countable parameter set A, let k be the differential field it generates. Then $S_1(A) \simeq S_1(k)$, and we have a bijection

$$S_1(k) \longleftrightarrow \mathrm{Spec}_\partial(k\{X\})$$

Each element of $\mathrm{Spec}_\partial(k\{X\})$ has the form $I(f)$ for $f \in k\{X\}$ an irreducible polynomial. The field k is countable, so $k\{X\}$ is countable. Therefore, $\mathrm{Spec}_\partial(k\{X\})$ is countable, so $S_1(k)$ is countable. ☐

Remark. A proof along the same lines shows that ACF_p is totally transcendental, using the bijection between types and prime ideals in polynomial rings.

We conclude this section with a remark on prime models of totally transcendental theories.

Definition. Given a set of parameters A, a model $M \supset A$ is *prime over* A if for any other model $N \supset A$, M admits an elementary embedding to N over A.

Theorem 10 (Morley, Ressayre, Shelah: see [Mar02]). *If T is a totally transcendental theory, then prime models exist over any set of parameters A. Moreover, prime models over A are unique up to isomorphism over A, and any type over A realized in a prime model is isolated.*

The proof of existence is fairly straightforward, using the abundance of isolated types in totally transcendental theories. Uniqueness is more complicated, involving the notion of nonforking independence.

In ACF_p, prime models over A are algebraic closures of the field generated by A. In DCF_0, we call a prime model over a differential field k a *differential closure* of k. A consequence of the isolation of types realized in the differential closure is that its field of constants is the algebraic closure of the field of constants of k. The notion of differential closure will not play a role in the rest of the proof, but we will use it in Section 6.

4.2. **Morley Rank.** In totally transcendental theories, types and formulas can be assigned dimension-like ranks. This gives us a useful lever over types and definable sets, which we will use to prove elimination of imaginaries in the next section.

In order to assign ranks to formulas with parameters from model, we will consider formulas with parameters from elementary extensions of it, so that the rank of a formula $\varphi(x, a)$ depends only on $\varphi(x, y)$ and $\mathrm{tp}(a)$, not the ambient model. We will also make use of the fact that if two tuples a and b in a model have the same type, there is a larger model with an automorphism sending $a \mapsto b$. See Appendix A for more on this.

Definition (Morley rank). For $\varphi(x)$ a formula with parameters in free variables x and α an ordinal, we define when $R(\varphi) \geq \alpha$ by recursion on α:

1. $R(\varphi) \geq 0$ if φ is consistent.
2. For β a limit ordinal, $R(\varphi) \geq \beta$ if for all $\alpha < \beta$, $R(\varphi) \geq \alpha$.
3. $R(\varphi) \geq \alpha + 1$ if there are formulas $\psi_i(x)$ for $i < \omega$, with parameters potentially in some larger model, such that
   (a) $R(\psi_i) \geq \alpha$ for each i.
   (b) $\psi_i(x) \vdash \varphi(x)$ for each i.
   (c) $\psi_i(x) \wedge \psi_j(x)$ is inconsistent for each $i \neq j$.

For definable sets, we are saying that a set has rank $\geq \alpha + 1$ if we can find infinitely many disjoint definable subsets of it of rank $\geq \alpha$. If $R(\varphi) \geq \alpha$ but $R(\varphi) \not\geq \alpha + 1$, we say $R(\varphi) = \alpha$. If φ is inconsistent we say $R(\varphi) = -1$. In either of these case we say that φ is ranked. If for all α, $R(\varphi) \geq \alpha$, we say that $R(\varphi) = \infty$, and that φ is unranked.

Lemma 7 (Basic Properties of Morley Rank). *In any theory, Morley rank of formulas has the following properties.*

1. *If $\varphi \vdash \psi$, $R(\varphi) \leq R(\psi)$*
2. *If φ is consistent, $R(\varphi) = 0$ if and only if φ has finitely many solutions. In this case we say that φ is an algebraic formula.*
3. $R(\varphi \vee \psi) = \max\{R(\varphi), R(\psi)\}$
4. $R(\varphi(x, a))$ *depends only on* $\varphi(x, y)$ *and* $\mathrm{tp}(a)$.

Proof sketch. (1) and (3) are proved using induction on rank. (2) is immediate from the definition of Morley rank. For (4), given two tuples a and b with the same type, we wish to prove by induction on α that $R(\varphi(x, a)) \geq \alpha \Rightarrow R(\varphi(x, b)) \geq \alpha$. The base and limit cases of the induction are immediate. For the successor case, we find formulas $\psi_i(x, c_i)$ witnessing $R(\varphi(x, a)) \geq \alpha + 1$. Then we choose a model M containing a, b, and the c_i, which has an automorphism taking $a \mapsto b$. This takes the $\psi_i(x, c_i)$ to formulas $\psi_i(x, d_i)$ witnessing $R(\varphi(x, b)) \geq \alpha + 1$. □

Theorem 11. *In a totally transcendental theory, every formula is ranked.*

This accounts for the name "totally transcendental." Morley defined a rank on types, rather than formulas, which he called "rank of transcendence" and called a theory totally transcendental if every type received a rank. We will see this rank on types later.

Proof. Consider the following set

$$\{(\varphi(x,y), q(y)) \mid \varphi(x,y) \text{ is a formula, } q(y) \in S(\emptyset) \text{ and if } \mathrm{tp}(a) = q \text{ then } R(\varphi(x,a)) < \infty\}$$

We define the rank of a pair $(\varphi(x,y), q(y))$ to be $R(\varphi(x,a))$ where $\mathrm{tp}(a) = q$. This is well defined by (4) in Lemma 7. Letting β be the supremum of the ranks of pairs in this set, we find that if φ is a formula and $R(\varphi) > \beta$, then $R(\varphi) = \infty$. If any φ is unranked, then $R(\varphi) \geq \beta + 2$, so we can find φ_0 and φ_1 such that for each i, $\varphi_i \vdash \varphi$, $R(\varphi_i) \geq \beta + 1$, and $\varphi_0 \wedge \varphi_1$ is inconsistent. Then each φ_i is unranked. By recursion, we build the forbidden configuration in (3) of Theorem 8. □

Remark. We can run this argument whenever there are ordinals $\alpha < \beta$ such that $R(\varphi) \geq \alpha \Rightarrow R(\varphi) \geq \beta$, and for some φ, $R(\varphi) \geq \beta$. This implies that in a totally transcendental theory, if $R(\varphi) = \beta$ for some β, then for every $\alpha \leq \beta$, some formula has rank α. For totally transcendental theories in a countable language, there are only countably many pairs $(\varphi(x,y), q(y))$, so this implies that ranks are countable ordinals.

In ACF_p, definable sets are Zariski constructible, and the Morley rank of a set is equal to its Krull dimension. In particular, all Morley ranks are finite ordinals. In DCF_0, infinite ordinals appear as Morley ranks. For instance, the formulas $x = x$ has rank ω.

Lemma 8 (Lemma/Definition)**.** *If $R(\varphi) = \alpha$, then there is a maximal $n < \omega$ such that there are $\psi_1, \ldots, \psi_n$ all of rank α, such that $\psi_i \wedge \psi_j$ is inconsistent for $i \neq j$, and $\psi_i \vdash \varphi$ for each i. We call this n the Morley degree of φ, denoted $D(\varphi)$.*

Proof. Given a formula ψ of rank α, we say that ψ is decomposable if there are ψ_0, ψ_1 both of rank α, such that $\psi_0 \vee \psi_1 \equiv \psi$ and $\psi_0 \wedge \psi_1$ is inconsistent. We construct by recursion a binary tree whose nodes are labelled by formulas of rank α, starting with φ at the root. Having constructed the tree to height n, we examine each leaf formula ψ. If it is not decomposable, it remains a leaf. Otherwise, we add two nodes succeeding it labelled by formulas ψ_0, ψ_1 witnessing a decomposition.

König's lemma implies that either the tree stops growing at some finite stage, so φ is equivalent to a finite disjunction of pairwise inconsistent indecomposable formulas of rank α, or the tree grows forever and has an infinite path.

$$\varphi = \psi_0, \psi_1, \psi_2, \ldots$$

Given such a path, the formulas $\psi_i \wedge \neg\psi_{i+1}$ are pairwise inconsistent, have rank α, and imply φ. By definition of Morley rank, this would imply that φ has rank at least $\alpha + 1$, a contradiction.

We conclude that φ is equivalent to a finite disjunction of pairwise inconsistent indecomposable formulas

$$\varphi \equiv \psi_1 \vee \cdots \vee \psi_n$$

Now given any pairwise inconsistent formulas of rank α, $\theta_1, \ldots, \theta_m$ with $\theta_j \vdash \varphi$, consider the formulas

$$\chi_{ij} = \psi_i \wedge \theta_j$$

Then for each j we have

$$\theta_j \equiv \chi_{1j} \vee \cdots \vee \chi_{nj}$$

so at least one of the χ_{ij} has rank α, whereas for each i at most one of the χ_{ij} has rank α, by the indecomposability of ψ_i. This implies that $m \leq n$, so n is the maximum we wanted. □

Corollary. *If φ and ψ both have Morley rank α, and $\varphi \wedge \psi$ is inconsistent, then*

$$D(\varphi \vee \psi) = D(\varphi) + D(\psi)$$

We can use our rank on formulas to define a rank on types.

Definition. For $p \in S(A)$ a type over some parameter set A, we define

$$R(p) = \inf\{R(\varphi) \mid \varphi \in p\}$$

We call a formula $\varphi \in p$ a *minimal formula* if $R(\varphi) = R(p)$, and φ has minimal Morley degree among formulas in p of rank $R(p)$.

We will now show that a type can be recovered from a minimal formula. Fix a set of parameters A, and let $L(A)$ denote the set of formulas with parameters from A. We say the a formula $\varphi \in L(A)$ is A-indecomposable if there do not exist formulas $\psi_0, \psi_1 \in L(A)$ such that $R(\psi_i) = R(\varphi)$ for each i, $\psi_0 \wedge \psi_1$ is inconsistent, and $\psi_0 \vee \psi_1 \equiv \varphi$.

Lemma 9. *If φ is a minimal formula of a type in $S(A)$, then φ is A-indecomposable.*

Proof. Suppose φ has a decomposition ψ_0, ψ_1. Then one of the ψ_i must be in the type, and has Morley degree lower than φ, a contradiction. □

Lemma 10. *Given an A-indecomposable formula φ, there is a unique type $p \in S(A)$ such that φ is a minimal formula for p.*

Proof. Let $\varphi(x)$ be an A-indecomposable formula of rank $\alpha \geq 0$. Consider the set of formulas

$$\Sigma = \{\psi(x) \in L(A) \mid R(\neg\psi \wedge \varphi) < \alpha\}$$

This set must be contained in any complete type with minimal formula φ, so it suffices to show that Σ is a complete type. Given finitely many formulas $\psi_1, \dots \psi_n \in \Sigma$ consider

$$\varphi \wedge \neg(\psi_1 \wedge \cdots \wedge \psi_n) = \bigvee_i (\varphi \wedge \neg\psi_i)$$

This is a finite disjunction of rank $< \alpha$ formulas, so has rank $< \alpha$. Therefore,

$$\varphi \wedge \psi_1 \wedge \cdots \wedge \psi_n$$

has rank α, and in particular is consistent. To show that Σ decides every formula in $L(A)$, suppose for a contradiction that neither ψ nor $\neg\psi$ are in Σ. Then both $\varphi \wedge \psi$ and $\varphi \wedge \neg\psi$ have rank α, contradicting the A-indecomposability of φ. □

To complete the picture, we say that two formulas φ, ψ in the same free variables of rank α are equivalent, $\varphi \sim_\alpha \psi$ if

$$R((\varphi \wedge \neg\psi) \vee (\neg\varphi \wedge \psi)) < \alpha$$

On the level of definable sets, this says that the symmetric difference of the two sets is small. Checking that this is an equivalence relation is a matter of using the fact that the disjunction of two rank $< \alpha$ formulas has rank $< \alpha$.

Lemma 11. *Any two minimal formulas of a rank α type are equivalent in the sense of $\sim_\alpha$. Any two equivalent A-indecomposable formulas of rank α are the minimal formula of the same type in $S(A)$.*

Proof. This follows from the description of the type in terms of a minimal formula in the previous lemma. □

Conclusion. *In any theory for which Morley rank is well founded, rank α types in $S(A)$ are in one to one correspondence with equivalence classes of rank α A-indecomposable formulas.*

This implies that if the language is countable and A is countable, there are only countably many types in $S(A)$, so the well foundedness of Morley rank is equivalent to our other definitions of totally transcendental.

We conclude this section with a definition which we will not use in the remainder of the proof, but has played an important role in the study of totally transcendental theories in general and DCF_0 in particular.

Definition. We say that a formula φ is *strongly minimal* if $R(\varphi) = 1$ and $D(\varphi) = 1$.

It is equivalent to say that φ has infinitely many solutions, and for any other formula ψ, either $\varphi \wedge \psi$ or $\varphi \wedge \neg\psi$ has finitely many solutions. On the level of definable sets, any definable subset of the set X defined by φ is either finite or cofinite in X. In ACF_p, irreducible algebraic curves are strongly minimal, and any strongly minimal set differs from an irreducible curve by finitely many points.

In [HS94], Hrushovski and Sokolović study strongly minimal sets in DCF_0, showing that they differ by a finite set of points from sets which have the structure of a *Zariski geometry* in the sense of [HZ96]. This allows a rough classification of strongly minimal sets into those which are in definable finite to finite correspondence with (nonorthogonal to) the field of constants, those whose structure is tied to that of an abelian group, and those which are geometrically trivial, meaning that algebraic relations between their elements are all essentially binary.

4.3. **Ideals and Types in Multiple Variables.** In this section, we return to DCF_0. We will show that types in multiple variables correspond to ideals, and study these ideals.

Theorem 12. *Let k be a differential field, viewed as a subfield of some model of DCF_0. For $p \in S_n(k)$, let $I(p)$ be the set*

$$\{f \in k\{X_1, \dots, X_n\} \mid p(\overline{x}) \vdash f(\overline{x}) = 0\}$$

Then

$$p \longmapsto I(p)$$

is a bijection

$$S_n(k) \longleftrightarrow \operatorname{Spec}_\partial(k\{X_1, \dots, X_n\})$$

Proof. The proof is identical to the case of types in a single variable. □

We will now study the algebraic side of this correspondence. While it has a certain explanatory power for constraints on the complexity of types in multiple variables, we will only need a small piece of this theory for elimination of imaginaries, so the greater part will be stated without proof. A key theorem is an analogue of the Hilbert basis theorem for differential polynomial rings. We cannot expect arbitrary differential ideals to be finitely generated in a direct sense. As a counterexample, consider the union of the infinite ascending chain of differential ideals

$$\langle (X^{(0)})^2 \rangle \subset \langle (X^{(0)})^2, (X^{(1)})^2 \rangle \subset \langle (X^{(0)})^2, (X^{(1)})^2, (X^{(2)})^2 \rangle \subset \dots$$

It will turn out that *radical* differential ideals are finitely generated in a more subtle way.

Definition. Given $f_1, \dots, f_r$ elements of a differential ring R, we let

$$\{f_1, \dots, f_r\} = \sqrt{\langle f_1, \dots, f_r \rangle}$$

We have seen that the radical of a differential ideal is a differential ideal, so this is the smallest radical differential ideal containing $f_1, \dots, f_r$.

Theorem 13 (Ritt-Raudenbush Basis Theorem, see [Kol73])**.** *Let k be a differential field, and $I \subset k\{X_1, \dots, X_n\}$ a radical differential ideal. Then there are $f_1, \dots, f_r \in I$ such that*

$$I = \{f_1, \dots, f_r\}$$

Equivalently, the ring $k\{X_1, \dots, X_n\}$ satisfies the ascending chain condition on radical differential ideals.

We now consider a differential polynomial analogue of primary decomposition in polynomial rings. This allows us to understand radical differential ideals in terms of prime components. We will actually use it in the other direction, to package finite sets of prime differential ideals as a single radical differential ideal.

Theorem 14 (Primary Decomposition of Radical Differential Ideals)**.** *Any radical differential ideal $I \subset k\{X_1, \dots, X_n\}$ can be expressed in a unique way as an intersection of finitely many prime differential ideals, none of which contain any other.*

The proof of existence is similar to the case of Noetherian rings, using the ascending chain condition and a certain lemma on products of radical differential ideals. The only part of the theorem we use in what follows is uniqueness, so we will prove that.

Proof of Uniqueness. This is a purely ring theoretic argument. Suppose

$$\mathfrak{p}_1 \cap \dots \cap \mathfrak{p}_m = \mathfrak{q}_1 \cap \dots \cap \mathfrak{q}_n$$

for prime differential ideals $\mathfrak{p}_i$ and $\mathfrak{q}_i$, with no ideal on either side containing any other. For each i we have the inclusion

$$\mathfrak{q}_1 \cap \dots \cap \mathfrak{q}_n \subset \mathfrak{p}_i$$

We wish to show that for some j, $\mathfrak{q}_j \subset \mathfrak{p}_i$. If not, we can choose $a_j \in \mathfrak{q}_j \setminus \mathfrak{p}_i$ for each j. Then the product of the a_j is in the intersection, but not in $\mathfrak{p}_i$, a contradiction. We conclude that $\mathfrak{q}_j \subset \mathfrak{p}_i$ for some j. Call this $j = f(i)$. Likewise, for any j, there is a $g(j)$ such that $\mathfrak{p}_{g(j)} \subset \mathfrak{q}_j$. Then for each i we have

$$\mathfrak{p}_{g(f(i))} \subset \mathfrak{q}_{f(i)} \subset \mathfrak{p}_i$$

As none of the $\mathfrak{p}_i$ contain each other, we have $g(f(i)) = i$, and $\mathfrak{p}_i = \mathfrak{q}_{f(i)}$. Likewise we have $g(g(j)) = j$ and $\mathfrak{q}_j = \mathfrak{p}_{g(j)}$. Therefore the $\mathfrak{p}_i$ and the $\mathfrak{q}_j$ are equal up to a permutation.

□

5. Elimination of Imaginaries

In this section we will use our understanding of definable sets and types in DCF_0 to prove that it admits elimination of imaginaries. We start by introducing some general model theoretic criteria for elimination of imaginaries. For the rest of the argument, instead of considering parameter sets in all differentially closed fields, we will fix in advance a large, saturated model $\mathcal{U} \models \mathsf{DCF}_0$ in which everything takes place. We then consider parameter sets $A \subset \mathcal{U}$ which are smaller than $\mathcal{U}$. Working in this setting entails no loss of generality, as the result we are proving is recorded in the complete theory DCF_0, and holds true in all its models.

Fixing a single structure $\mathcal{U}$ lets us refer to its automorphism, which usefully characterize certain logical notions. We will use the following property of saturated models.

Proposition. *For a, b tuples in $\mathcal{U}$, $A \subset \mathcal{U}$ a parameter set smaller than $\mathcal{U}$, the following are equivalent:*

1. $\mathrm{tp}(a/A) = \mathrm{tp}(b/A)$.
2. *There is an automorphism f of $\mathcal{U}$ fixing A pointwise and sending $a \mapsto b$.*

For more one these matters, see Appendix A.

5.1. **Canonical Bases.**

Definition. Let $\varphi(x, a)$ be a formula, defining a subset $\varphi(\mathcal{U})$. We say that a set B is a *canonical base* for $\varphi(x, a)$, or for the definable set $\varphi(\mathcal{U})$, if the automorphisms fixing B pointwise are exactly the automorphisms fixing $\varphi(\mathcal{U})$ setwise. In other words

$$f(b) = b \text{ for all } b \in B \iff \varphi(x, a) \equiv \varphi(x, f(a))$$

Canonical bases occur in Galois theory. Given a finite subset of a field $Z = \{a_1, \ldots, a_n\}$, a canonical base for Z is the set of symmetric polynomials in the a_i, $\{s_1(a_1, \ldots, a_n), \ldots, s_n(a_1, \ldots, a_n)\}$. These are the coefficients of the unique monic polynomial with simple zeros Z. This implies that an automorphism of the ambient field permutes Z if and only if it fixes the coefficients of this polynomial. In Section 6 we will see an analogous construction from the Galois theory of linear differential equations, a canonical base for a finite dimensional vector space over the constants.

Remark. A way of defining this which is agnostic to the model $\mathcal{U}$ is that B is a canonical base for $\varphi(x, a)$ if $\varphi(x, a)$ is equivalent to a formula with parameters from B, and each element of B is definable in terms of the relation $\varphi(x, a)$.

The main result of this section will allow us to move from canonical bases in this sense to finite tuples which can be obtained by applying definable functions to the parameters of the formula. This is in the spirit of symmetric polynomials and moves us closer to elimination of imaginaries.

Lemma 12 (Existence of canonical parameters)**.** *If $\varphi(x, a)$ has a canonical base B, then there is a formula $\psi(x, y)$ and a tuple b from B such that b is the unique tuple satisfying*

$$\varphi(x, a) \equiv \psi(x, b)$$

This implies that:

1. *$\{b\}$ is a canonical base for $\varphi(x, a)$*
2. *b is definable from a, it is the image of a under a definable function.*

We call such a b a canonical parameter for $\varphi(x, a)$.

Proof. Consider the set Γ of formulas $\theta(x, b)$ where b is a tuple from B and $\varphi(x, a) \vdash \theta(x, b)$. I claim that $\Gamma \vdash \varphi(x, a)$. If $\Gamma \not\vdash \varphi(x, a)$, by the saturation properties of $\mathcal{U}$ there is a c_0 in $\mathcal{U}$ such that $\Gamma(c_0) \wedge \neg\varphi(c_0, a)$ holds. On the other hand, for any c_1 in $\mathcal{U}$ such that $\varphi(c_1, a)$ holds, $\mathrm{tp}(c_1/B) \neq \mathrm{tp}(c_0/B)$. Therefore, $\mathrm{tp}(c_0/B) \vdash \neg\varphi(x, a)$, so for some formula $\eta(x, b) \in \mathrm{tp}(c_0/B)$, we have

$$\eta(x, b) \vdash \neg\varphi(x, a)$$

which in turn implies

$$\varphi(x,a) \vdash \neg\eta(x,b)$$

so $\neg\eta(x,b) \in \Gamma \subset \text{tp}(c_0/B)$, a contradiction.

We conclude that $\Gamma \vdash \varphi(x,a)$, so by compactness, some $\theta(x,b) \in \Gamma$ is equivalent to $\varphi(x,a)$. If f is an automorphism moving b to $b_1 \neq b$, then $\varphi(x,f(a)) \not\equiv \varphi(x,a)$. Letting $p = \text{tp}(b/\emptyset)$, we have

$$p(y) \cup \{y \neq b\} \vdash \varphi(x,a) \not\leftrightarrow \theta(x,y)$$

Consequently, for some $\chi(y) \in p$,

$$\chi(y) \wedge y \neq b \vdash \varphi(x,a) \not\leftrightarrow \theta(x,y)$$

Letting $\psi(x,y) = \theta(x,y) \wedge \chi(y)$, it follows that

$$\psi(x,b) \equiv \varphi(x,a)$$

and b is unique as such, provided that $\varphi(x,a)$ is consistent. If $\varphi(x,a)$ is inconsistent, we can take b to be the empty tuple, so the lemma is true in this case as well. $\square$

5.2. **Imaginary Elements.** At this point we can introduce imaginaries formally. Given a definable set X and a definable equivalence relation E on X, we call equivalence classes a/E where $a \in X$ *imaginary elements.*

Definable sets can be viewed as imaginary elements. Given a formula $\varphi(x,y)$, define the equivalence relation

$$yEz \iff (\forall x)(\varphi(x,y) \leftrightarrow \varphi(x,z))$$

Then the imaginary a/E can be viewed as a representative for the set defined by $\varphi(x,a)$. Automorphisms fix $\varphi(\mathfrak{U})$ setwise if and only if they fix a/E. We can say in this sense that any definable set has a canonical parameter, just that this canonical parameter may be an imaginary element. This is one of the roles played by imaginary elements in [She78].

The main theorem of this section shows that the existence of canonical bases for formulas is equivalent to the existence of definable functions representing quotients by definable equivalence relations. Such functions provide a definable correspondence between imaginary elements and ordinary tuples.

Theorem 15. *Let T be a theory such that at least two distinct elements are definable (for example, any theory of a field.) The following are equivalent.*

(1) *In some/any saturated model $\mathfrak{U}$ of T, each formula has a canonical base.*
(2) *For A a set of parameters, given any A-definable set X and A-definable equivalence relation E on X, there is an A-definable surjective function $f : X \to Y$ such that for $a, b \in X$*

$$aEb \iff f(a) = f(b)$$

Proof. Suppose T satisfies (2), and let $\varphi(x,a)$ be a formula with parameters from a homogeneous model $\mathfrak{U}$. Consider the $\emptyset$-definable equivalence relation aEb if $\varphi(x,a) \equiv \varphi(x,b)$. By (2) there is a $\emptyset$-definable function f such that aEb if and only if $\varphi(x,a) \equiv \varphi(x,b)$. Consider $f(a)$. Any automorphism of $\mathfrak{U}$ fixing $\varphi(x,a)$ fixes setwise the E-equivalence class of a, so fixes $f(a)$. On the other hand, any automorphism fixing $f(a)$ fixes setwise its fiber, the E-equivalence class of a, and so fixes $\varphi(x,a)$ setwise. Therefore, $\{f(a)\}$ is a canonical base for $\varphi(x,a)$.

Suppose now that X satisfies (1) for some saturated model $\mathfrak{U}$, and consider a definable set X defined by $\varphi(x,a)$, and equivalence relation E defined by $\psi(x,y,a)$. What we want to prove is a property of $\text{tp}(a)$, so we may assume a is in $\mathfrak{U}$. Now working in $\mathfrak{U}$, for any $b \in X$, there is a canonical base for $\psi(x,a,b)$, so by Lemma 12 there is a formula θ and a tuple c such that c is the unique tuple satisfying

$$\psi(x,a,b) \equiv \theta(x,c)$$

For fixed θ, the existence of such a c is a definable property of b. For any element of X, some θ works, so by compactness there are finitely many formulas $\theta_1, \ldots, \theta_r$ such that one of them works for every $b \in X$. We construct $\theta(x,y)$ such that the entries of y encode using the two definable elements an $i \leq r$ and a parameter w for θ_i. The formula $\theta(x,y)$ says "$\theta_i(x,w)$ and i is least such that θ_i works for this value of x." Then the formula

$$\varphi(x,a) \wedge \forall z\,[\psi(z,a,y) \leftrightarrow \theta(z,y)]$$

defines a function $f : x \mapsto y$ such that $f(x_1) = f(x_2)$ if and only if $x_1 E x_2$. □

Note that while (1) refers to a certain class of models, (2) is recorded in the theory and holds in all models. If a theory has this property, we say that it *eliminates imaginaries.* The specific property defined here is often called *uniform elimination of imaginaries* due to the uniformity afforded by a single definable function for the entire domain of the equivalence relation. In theories with at least two definable elements, there is no distinction between uniform elimination of imaginaries and a more general definition, so as we are considering theories of fields we will not make this distinction.

We will now make some observations on what was used in the above proof, in order to extract a criterion for elimination of imaginaries. To prove (2) $\implies$ (1), we only considered $\emptyset$-definable equivalence relations whose domain is a Cartesian power of the model. In fact, given such an equivalence relation E, if every equivalence class of E has a canonical base, then the proof of (1) $\implies$ (2) can be used to show that there is a definable function f which represents the quotient by E. From this, the proof that (2) $\implies$ (1) can be used to conclude that (1) holds for all formulas. Therefore we have the following criterion.

Lemma 13. *If T has at least two definable elements, to prove that T eliminates imaginaries it suffices to prove that for any $\emptyset$-definable equivalence E relation on a cartesian power of a saturated model $\mathfrak{U}$, each equivalence class has a canonical base.*

In a totally transcendental theory, we can push this further. For a set P of types in $S(\mathfrak{U})$, we say that B is a canonical base for P if the automorphisms fixing B pointwise are exactly those permuting P. We say that two types $p, q \in S(\mathfrak{U})$ are *conjugate* if an automorphism takes one to the other.

Lemma 14. *Suppose T is a totally transcendental theory with at least two definable elements. Then if every finite set of conjugate types $P \subset S(\mathfrak{U})$ has a canonical base (for some $\mathfrak{U}$), T eliminates imaginaries.*

Proof. We prove that any equivalence class for a $\emptyset$-definable equivalence relation has a canonical base. Let Ea be such an equivalence class. The formula xEa can be written as a disjunction of formulas $\psi_1 \vee \cdots \vee \psi_r$ of Morley rank equal to that of xEa and Morley degree 1. By the saturation properties of $\mathfrak{U}$, we may assume that the parameters of these formulas are in $\mathfrak{U}$. Each formula ψ_i is the minimal formula of a type $p_i \in S(\mathfrak{U})$. We can divide these types $p_1, \ldots, p_r$ into conjugacy classes $P_1, \ldots, P_k$. For each i, let B_i be a canonical base for P_i, and let $B = B_1 \cup \cdots \cup B_k$. We will show that B is a canonical base for xEa. The types p_i are the types of maximal rank containing xEa, so are permuted by any automorphism fixing Ea setwise. Therefore, such automorphisms fix B pointwise. Conversely, any automorphism fixing B pointwise permutes the types p_i. For f such an automorphism, the formula $xEf(a)$ must be contained in each type p_i, so these contain $xEa \wedge xEf(a)$. Therefore, $xEf(a) \wedge xEa$ is consistent, so because E is an equivalence relation, $aEf(a)$, or in other words $Ea = Ef(a)$. Thus f fixes Ea setwise. □

5.3. **Fields of Definition.** In this section, we will complete the proof of elimination of imaginaries in DCF_0 by finding the required canonical bases. The same strategy will work for ACF_p, so we will do this as well. In both cases, the existence of canonical bases follows from Weil's theory of fields of definition.

Definition. Let K be a field, and consider a polynomial ring

$$S = K[x_\alpha]_{\alpha \in A}$$

over K in some possibly infinite set of variables $\{x_\alpha\}$. Given an ideal $I \subset S$, a subfield $k \subset K$ is called a *field of definition* for I if I can be generated by polynomials with coefficients in k.

Theorem 16 (Weil). *Any ideal $I \subset S$ has a unique minimal field of definition, contained in every other field of definition.*

Proof. The proof is based on the principle that the dimension of the solution set of a linear equation does not change when we extend scalars. The K-vector space S/I is spanned by the images of monomials, so we can choose once and for all a basis of monomials $\{x^\mu \mid \mu \in M\}$. Then for any multi-index ν, we can write

$$x^\nu = \sum_{\mu \in M} \lambda_{\nu\mu} x^\mu$$

Let k be the field generated by the $\lambda_{\nu\mu}$. Now let $\{f_j\}_{j\in J}$ generate I. These generators are linear combinations of monomials

$$f_j = \sum_\nu \alpha_{j\nu} x^\nu$$

which are trivial in S/I. In terms of the monomial basis, we have for fixed j

$$0 = \sum_\nu \sum_\mu \lambda_{\nu\mu}\alpha_{j\nu}x^\mu$$

Therefore, for each fixed μ,

$$\sum_\nu \lambda_{\nu\mu}\alpha_{j\nu} = 0$$

We can view this as a product of a row vector and a column vector

$$0 = \left[\lambda_{\nu\mu}\right]\left[\alpha_{j\nu}\right]$$

for fixed j and μ. Let Λ_μ be this row vector of $\lambda_{\nu\mu}$ The kernel of Λ_μ as a map $K^n \to K$ is the tensor product of K and its kernel in k^n, so we can express the vector of the $\alpha_{j\nu}$ as a K-linear combination of elements of the kernel of Λ_μ over k. This expresses f_j as a K-linear combination of elements of $I \cap k[x_\alpha]$, so I can be defined over k.

Suppose conversely that $L \subset K$ is a subfield such that there are generators $\{h_p\}_{p\in P}$ for I with coefficients in L. For any ν we can write

$$x^\nu - \sum_{\mu\in M}\lambda_{\nu\mu}x^\mu = \sum_l g_l h_l$$

for some $g_l \in S$. We view this equation as a system of linear equations whose variables are the $\lambda_{\nu\mu}$ and the coefficients of the g_l. This equation has coefficients in L, coming from the coefficients of the h_l. Because this linear equation has solutions over K, it has solutions over L. On the other hand, the $\lambda_{\nu\mu}$ are uniquely determined by being solutions to such an equation because the x^μ for $\mu \in M$ are a basis for S/I. Therefore, each $\lambda_{\nu\mu} \in L$, so $k \subset L$. □

Corollary (of the proof)**.** *An automorphism of K fixes I setwise if and only if it fixes the minimal field of definition k pointwise.*

Theorem 17. DCF_0 *eliminates imaginaries.*

Proof. Let $\{p_1,\dots,p_n\}$ be a finite conjugate set of types in $S(\mathfrak{U})$ for some large saturated model $\mathfrak{U}$ of DCF_0. These correspond to prime differential ideals

$$\mathfrak{p}_1,\dots,\mathfrak{p}_n \in \operatorname{Spec}_\partial(\mathfrak{U}\{X_1,\dots,X_n\})$$

which are conjugate under the automorphism group of $\mathfrak{U}$. Now consider the radical differential ideal

$$I = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_n$$

This is the unique primary decomposition of the radical differential ideal I. Viewing I as an ideal in a polynomial ring and forgetting the differential structure, we have a field of definition k. Then given any $f \in \operatorname{Aut}_\partial(\mathfrak{U})$, f is in particular an automorphism of fields, so the following are equivalent:

(1) f fixes k pointwise.
(2) f fixes I setwise.
(3) f permutes the ideals $\mathfrak{p}_i$.
(4) f permutes the types p_i.

We conclude by Lemma 14. □

Remark. An analogous argument shows that ACF_p eliminates imaginaries.

To summarize the proof, we wished to show that a quotient by a definable equivalence relation can be represented by a definable function. Using compactness, this uniform statement can be deduced from the existence of canonical parameters for individual equivalence classes. This can in turn be deduced from the existence of a canonical base for each class, using further compactness principles. To find such a canonical base, we used the fact that in a totally transcendental theory, each definable set has a finite set of maximal rank types attached to it. These types correspond prime differential ideals in a ring of differential polynomials, which we can package them together by taking their intersection. A minimal field of definition for this ideal is then our canonical base.

6. Examples of Eliminating Imaginaries

Our strategy of proof in the previous section was indirect, relying on compactness principles in several places. As a result, it does not give us much information about how to find definable functions witnessing elimination of imaginaries for particular equivalence relations. In this section we will see some examples of how this can work.

6.1. Basic Examples.

Consider the equivalence relation of differing by a constant

$$xEy \iff (\exists z)(\partial z = 0 \wedge x = y + z)$$

Here xEy if and only if $\partial x = \partial y$. Therefore, the derivative $x \mapsto \partial x$ witnesses elimination of imaginaries for this equivalence relation.

Now consider the equivalence relation of differing by a nonzero constant multiple

$$xEy \iff (\exists z)(\partial z = 0 \wedge z \neq 0 \wedge x = y \cdot z)$$

The logarithmic derivative

$$\partial \log : x \mapsto \frac{\partial x}{x}$$

witnesses elimination of imaginaries for this equivalence relation. (Strictly speaking, we need to have a special case for $x = 0$, which is its own equivalence class.) If $x = c \cdot y$, where c is constant, then

$$\frac{\partial y}{y} = \frac{c\partial x}{cx} = \frac{\partial x}{x}$$

Conversely, if x and y are nonzero

$$\partial \log x = \partial \log y$$

then

$$\partial \left[\frac{y}{x}\right] = \frac{\partial y}{x} - \frac{y \cdot \partial x}{x^2} = \frac{y}{x}\left[\frac{\partial y}{y} - \frac{\partial x}{x}\right] = 0$$

so x and y differ by a constant multiple.

6.2. Canonical Parameters for Vector Spaces.

In a model K of DCF_0, if we are given elements $y_1, \dots, y_n \in K$, we can consider the vector space over the constants C spanned by the y_i. We define the *Wronskian* of $y_1, \dots, y_n$ to be the differential polynomial

$$\mathrm{wr}(y_1, \dots, y_n) = \det \begin{bmatrix} y_1 & \cdots & y_n \\ \vdots & & \vdots \\ \partial^{n-1} y_1 & \dots & \partial^{n-1} y_n \end{bmatrix}$$

This detects when the vector space is n-dimensional.

Proposition ([vdPS03] Lemma 1.12)**.** *Given $y_1, \dots, y_n \in K$ a differential field,* $\mathrm{wr}(y_1, \dots, y_n) = 0$ *if and only if $y_1, \dots, y_n$ are linearly dependent over the constants.*

On the locus of $(y_1, \dots, y_n)$ where the Wronskian does not vanish, we can define an equivalence relation

$$(y_1, \dots, y_n)E(z_1, \dots, z_n) \iff \mathrm{span}_C\{y_1, \dots, y_n\} = \mathrm{span}_C\{z_1, \dots, z_n\}$$

To eliminate imaginaries here, we need canonical parameters for vector spaces. These will be coefficients of a linear differential equation whose solution space is the given vector space, analogous to symmetric polynomials, which are coefficients of a polynomial coding a finite set.

We write the desired linear differential equation as

$$\partial^n y + a_{n-1}\partial^{n-1} + \cdots + a_0 y = 0$$

where the a_i are to be determined. This should have solutions of the form

$$y = t_1 y_1 + \cdots + t_n y_n$$

where t_i are arbitrary constants. Plugging this in to the equation, we obtain

$$[t_1\partial^n y_1 + \cdots + t_n\partial^n y_n] + a_{n-1}\left[t_1\partial^{n-1}y_1 + \cdots + t_n\partial^{n-1}y_n\right] + \cdots + a_0\left[t_1y_1 + \cdots + t_ny_n\right] = 0$$

Comparing the coefficients on the indeterminants, we get a system of linear equations

$$\begin{bmatrix}\partial^n y_1\\ \vdots \\ \partial^n y_n\end{bmatrix} = -\begin{bmatrix} y_1 & \dots & \partial^{n-1}y_1\\ \vdots & & \vdots\\ y_n & \dots & \partial^{n-1}y_n\end{bmatrix}\begin{bmatrix}a_0\\ \vdots\\ a_{n-1}\end{bmatrix}$$

The square matrix appearing here has determinant $(-1)^n\mathrm{wr}(y_1,\dots,y_n) \neq 0$, so it is invertible. We can therefore write

$$\begin{bmatrix}a_0\\ \vdots\\ a_{n-1}\end{bmatrix} = -\begin{bmatrix} y_1 & \dots & \partial^{n-1}y_1\\ \vdots & & \vdots\\ y_n & \dots & \partial^{n-1}y_n\end{bmatrix}^{-1}\begin{bmatrix}\partial^n y_1\\ \vdots \\ \partial^n y_n\end{bmatrix}$$

This expresses the a_i as differential rational functions of the y_i. The map $(y_i) \mapsto (a_i)$ given by these functions witnesses elimination of imaginaries for this equivalence relation.

In the Galois theory of linear differential equations, the construction here can be used to find a Picard-Vessiot extension of differential fields whose Galois group is GL_n, see [vdPS03] Exercise 1.35. This can be compared to the use of symmetric polynomials to construct a finite field extension with Galois group S_n.

6.3. **The Schwarzian Derivative.** Consider the action of $\mathrm{PGL}_2 \curvearrowright \mathbb{P}^1$ by Möbius transformations.

$$\begin{bmatrix}a & b\\ c & d\end{bmatrix}\cdot[u:v] = [au+bv : cu+dv]$$

Viewing elements of our differential field K as functions to $\mathbb{P}^1$, we can define an equivalence relation

$$yEz \iff \exists \begin{bmatrix}a & b\\ c & d\end{bmatrix} \in \mathrm{GL}_2(C) \text{ such that } z = \frac{ay+b}{cy+d}$$

expressing that y and z are differ by postcomposition by an element $\mathrm{PGL}_2(C)$. We view this as an action of $\mathrm{PGL}_2(C)$ on the nonconstant elements of K, so that the expression $\frac{ay+b}{cy+d}$ is always well defined. We will show that elimination of imaginaries for this equivalence relation is witnessed by the *Schwarzian derivative*

$$Sy = \partial\left[\frac{\partial^2 y}{\partial y}\right] - \frac{1}{2}\left[\frac{\partial^2 y}{\partial y}\right]^2$$

In other words, we prove that for nonconstant y, z, $Sy = Sz$ if and only if

$$z = \frac{ay+b}{cy+d} \text{ for some } \begin{bmatrix}a & b\\ c & d\end{bmatrix} \in \mathrm{GL}_2(C)$$

We will show that Sy is interdefinable with the PGL_2-orbit of y in two steps. First we find a 2 dimensional vector space over the constants interdefinable with the PGL_2-orbit, then look at the canonical parameters of this vector space.

Given a 2-dimensional C vector space $V \subset K$, we obtain a PGL_2-orbit

$$\{v/u \mid u, v \text{ is a basis for } V\}$$

For a given orbit $\mathrm{PGL}_2 \cdot y$, there are many vector spaces which have that orbit as their set of basis ratios, all of the form

$$\mathrm{span}_C\{u, y\cdot u\}$$

for some element $u \in K$. To pick out a specific vector space, we look at its canonical parameters. Using the formula from the previous section, from $V = \mathrm{span}_C\{u, v\}$ we get the parameters

$$\begin{bmatrix}a_0\\ a_1\end{bmatrix} = -\begin{bmatrix}u & \partial u\\ v & \partial v\end{bmatrix}^{-1}\begin{bmatrix}\partial^2 u\\ \partial^2 v\end{bmatrix} = \frac{-1}{u\partial v - v\partial u}\begin{bmatrix}\partial v & -\partial u\\ -v & u\end{bmatrix}\begin{bmatrix}\partial^2 u\\ \partial^2 v\end{bmatrix}$$

This comes out to

$$a_0 = \frac{\partial^2 v\partial u - \partial^2 u\partial v}{u\partial v - v\partial u}$$

$$a_1 = \frac{v\partial^2 u - u\partial^2 v}{u\partial v - v\partial u} = -\partial \log(\mathrm{wr}(u, v))$$

Taking $v = y \cdot u$, we get

$$a_1 = -\partial \log(u \cdot \partial(y \cdot u) - y \cdot u \cdot \partial u) = -\partial \log(u^2 \cdot \partial y + y \cdot u \cdot \partial u - y \cdot u \cdot \partial u) = -\partial \log(u^2 \cdot \partial y)$$

Therefore, the constraint $a_1 = 0$ forces u to be a constant multiple of $(\partial y)^{-1/2}$, and uniquely determines V. This V is then interdefinable with $\mathrm{PGL}_2 \cdot y$, viewing both objects as imaginaries. To obtain an element interdefinable with $\mathrm{PGL}_2 \cdot y$, we look to the other canonical parameter a_0. We know that $u = (\partial y)^{-1/2}$ satisfies a linear differential equation

$$\partial^2 u + a_0 u = 0$$

To find a_0, we compute

$$\partial u = -\frac{\partial^2 y}{2(\partial y)^{\frac{3}{2}}} = -\frac{1}{2} \cdot \frac{\partial^2 y}{\partial y} \cdot u$$

and so

$$\partial^2 u = -\frac{1}{2}\left(\partial\left[\frac{\partial^2 y}{\partial y}\right] - \frac{1}{2}\left[\frac{\partial^2 y}{\partial y}\right]^2\right) u$$

Therefore, $a_0 = \frac{1}{2} Sy$. This shows that Sy is interdefinable with $\mathrm{PGL}_2 \cdot y$ in a uniform way, which implies the property we want.

6.4. **Algebraic Group Actions and Generalized Schwarzians.** In this section, we consider definable equivalence relations of a particular form. Let $G \curvearrowright X$ be the action of an algebraic group on an algebraic variety, all defined over the field of constants C. We then have a group action

$$G(C) \curvearrowright X(K)$$

which gives us an equivalence relation on $X(K)$

$$xEy \leftrightarrow (\exists g \in G(C))(g \cdot x = y)$$

Elimination of imaginaries implies that there is a piecewise differential rational function

$$f : X \to Y$$

where Y is Kolchin constructible, such that $f(x) = f(y)$ if and only if x and y are in the same $G(C)$ orbit. All of our previous examples can be viewed as special cases of this one, with groups G_a, G_m, GL_n and PGL_2. We call such a function f a *generalized Schwarzian derivative* of the group action.

Using the mechanism of differential closure, we can extend this result from differential fields to more general differential rings.

Proposition. *Suppose R is a differential ring whose ring of constants C is an algebraically closed field of characteristic* 0*, and $G \curvearrowright X$ is an action of an algebraic group on an algebraic variety defined over C. Then there is a piecewise differential rational map*

$$f : X(R) \to K(R)^n$$

defined using parameters from C, such that given $x, y \in X(R)$, $f(x) = f(y)$ if and only if there is a $g \in G(C)$ with $g \cdot x = y$.

Proof. Let L be the differential closure of $K(R)$, as described in Section 4. Because C is algebraically closed, the field of constants of L is also C. By elimination of imaginaries, there is a function

$$f : X(L) \to L^n$$

which has the claimed property. This same function works for R as well, because it is defined over C and the equivalence relations

$$f(x) = f(y)$$

and

$$(\exists g \in G(C))(g \cdot x = y)$$

are absolute between R and L. $\square$

Let $G \curvearrowright X$ be an algebraic group action defined over an algebraically closed field k. The set of rational maps $\mathbb{P}^1 \dashrightarrow X$ can be identified with the set of $k(t)$ points of X. We can give $k(t)$ the structure of a differential field with constants k, so as a consequence of the theorem, rational maps $\mathbb{P}^1 \dashrightarrow X$ can be classified up to the action of $G(k)$ using finitely many invariant differential rational functions of their coordinates.

The only role the derivation plays in defining the imaginaries here is in defining the field of constants. On the other hand, we need the derivation to eliminate them; the theory of an algebraically closed field with a distinguished algebraically closed subfield of constants does not eliminate imaginaries. The obstruction is essentially what is considered in this section, where constant points of an algebraic group group act on a variety, as shown in [Pil07].

In [Sca18], Scanlon uses the generalized Schwarzian in the following construction. Given $\Gamma \subset G(\mathbb{C})$ a Zariski dense discrete subgroup, such that the quotient $\Gamma \setminus X(\mathbb{C})$ is a variety $Y(\mathbb{C})$, letting $\pi : X(\mathbb{C}) \to Y(\mathbb{C})$ be the quotient map, we have a well defined operator $f \mapsto \chi(\pi^{-1} \circ f)$ acting on $Y(\mathcal{O})$, where $\mathcal{O}$ is some ring of analytic functions and χ is the generalized Schwarzian of $G \curvearrowright X$. Under a fairly general hypothesis concerning the quotient map, this is shown to be a differentially constructible map, or in other words, piecewise differential-rational. Examples include the logarithmic derivative $\partial \log$ and an operator given by the invariant differential on an elliptic curve which gives us the differential equation satisfied by the associated elliptic function.

7. Questions

Our proof of elimination of imaginaries does not provide us with a method of computing the function which realizes a quotient. In the case of algebraically closed fields, a somewhat more direct proof is available, due to Jan Holly in [Hol93]. The idea is essentially to find canonical parameters by first finding them "up to finite ambiguity" by choosing representatives which are algebraic over the input parameters, then using symmetric polynomials to turn these into genuine canonical parameters. Finding a more direct proof of elimination of imaginaries in DCF_0 along similar lines could be a first step to finding systematic ways of computing the functions which witness elimination of imaginaries.

The structure of strongly minimal sets in DCF_0 has played a major role in its study, following [HS94]. A major problem in this area is finding methods to determine whether differential equations of a certain form define strongly minimal sets. Examples of equations known to be strongly minimal include any order 1 linear differential equation, the equation

$$\partial^2 y = \frac{\partial y}{y}$$

and the Painlevé equation:

$$\partial^2 y = 6y + t$$

where $\partial t = 1$. For the latter two, see [MMP96].

As for the model theory of pure fields, in [Mac71], Macintyre proved that any infinite field with a totally transcendental theory is algebraically closed. The proof uses a combination of results on totally transcendental groups and Galois theory, reducing the problem to one of showing that a totally transcendental field can have neither Kummer extensions nor Artin-Schreier extensions. Cherlin and Shelah extended this result to the more general class of superstable fields in [CS80] with a more extensive use of the machinery of rank. A description of the still more general class of stable fields remains an open problem.

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Appendix A. General Model Theory

Here I sketch out, mostly without proof, the notions and tools from model theory that we use throughout the paper. Good references for this include [Mar02] and [Sac10].

First Order Languages and Compactness. Model theory is the study of formal languages and their interpretations, along with the associated notions of definability, axiomatizability, interpretability, expressive power, and classification of possible variation among models. In our context, we will consider first order languages.

Definition. A *signature* is a collection of constant, function, and relation symbols. Each function and relation symbol is assigned an arity, the number of arguments it takes.

The pertinent signature here is that of differential rings, which includes constant symbols 0 and 1, binary functions $+$ for addition, $\cdot$ for multiplication, and $-$ for subtraction, as well as a unary function symbol ∂ for a derivation. The signature has no relation symbols, although the equality symbol $=$ is included as a logical symbol in all first order languages.

A signature generates a first order language in the following way. *Terms* are constructed recursively by applying function symbols, starting with free variables and constants. *Atomic formulas* are formed by applying relation symbols (and equality) to terms. Formulas in general are built out of atomic formulas using boolean logical operations $\wedge, \vee, \neg, \to$ as well as quantification $\exists$ and $\forall$ over free variables. A *sentence* is a formula in which all variables are bound by quantifiers. For a signature τ, $L_{\omega\omega}(\tau)$ denotes the first order language generated by τ. Elsewhere, we will refer to the language as a whole as just L, and not refer to the signature.

First order languages are interpreted in structures defined as follows.

Definition. For a signature τ, a τ-structure (also called an L-structure) consists of an underlying set X together with an interpretation of each symbol of τ on X. Constant symbols are interpreted as elements, n-ary function symbols as functions $X^n \to X$, and n-ary relation symbols as subsets of X^n.

Given a τ-structure M, we can by recursion on the structure of a formula $\varphi(\overline{x})$ define when

$$M \models \varphi(\overline{a})$$

for $\overline{a}$ an assignment of elements of M to the free variables of φ, meaning that φ is true in M of the elements $\overline{a}$. For a sentence, there are no free variables and we can simply write $M \models \varphi$.

By a *theory* we mean a set of sentences in the language $L_{\omega\omega}(\tau)$. We say M is a model of a theory T, $M \models T$, if for every $\varphi \in T$, $M \models \varphi$. A theory is said to be consistent if it has a model. For a theory T and a sentence φ, we say that T entails φ, $T \vdash \varphi$, if every model of T is a model of φ. Equivalently, $T + \neg\varphi$ is inconsistent. We say that a theory T is complete if for any sentence φ, either $T \vdash \varphi$ or $T \vdash \neg\varphi$. For any structure M, we have a complete theory $\mathrm{Th}(M)$, the set of sentences true in M.

A foundational theorem in the study of first order logic is the following, which implies that consistency and entailment have a finitary character.

Theorem 18 (Compactness)**.** *Let T be a theory in a first order language.*

(1) *If every finite subset of T is consistent, then T is consistent.*
(2) *If $T \vdash \varphi$, then for some finite subset $\Sigma \subset T$, $\Sigma \vdash \varphi$.*

Given a structure M and a set A of elements of M, we denote by $L(A)$ the expanded language obtained by adding constant symbols naming each element of A. In this way, M and any structure extending it come with a natural interpretation of $L(A)$. We refer to the elements of A as parameters in this context.

Given structures M and N, we write $M \subset N$ if the underlying set of M is a subset of the underlying set of N and the interpretation of the functions, relations, and constants in N extend those in M. Given any subset $A \subset M$ of a structure M, there is a smallest substructure containing it, denoted $\langle A \rangle$. Its elements are denoted by terms in the language $L(A)$.

We write $M \preceq N$ and say that M is an *elementary substructure* of N, N is an *elementary extension* of M, if in addition for any formula $\varphi(\overline{x})$ of the language and tuple a from M,

$$M \models \varphi(a) \iff N \models \varphi(a)$$

In other words, M and N agree on the truth of all formulas describing elements of M. A *elementary map* is similarly a partial map between structures which preserves the truth of all formulas. An embedding of one structure into another which is an elementary map is called a *elementary embedding*.

Given a structure M, we can consider the language $L(M)$, naming all of its elements. The theory of M in this expanded language is called the *elementary diagram* of M, denoted $E(M)$. The quantifier free part of this is called the atomic diagram of M, denoted $D(M)$. Models of $D(M)$ can be identified with structures N containing M as a substructure, while models of $E(M)$ can be identified with elementary extensions of M.

A subset $X \subset M^n$ if called a *definable set* if for some formula $\varphi(\overline{x}) \in L(M)$ we have

$$X = \{\overline{a} \in M^n \mid M \models \varphi(\overline{a})\}$$

Boolean operations on formulas correspond to boolean operations on definable sets, and existential quantification corresponds to projection.

Types, Saturation, and Automorphisms. A type is a consistent set of formulas in some free variables. Types in n free variables are called n-types. We will deal mostly with complete types, which make a decision about every formula in the language with a given list of free variables. Given a tuple a of length n in a structure M, we have a complete n-type $\mathrm{tp}(a)$, the set of formulas true of a in M. By a type over a complete theory T we mean a type consistent with T. We denote by $S_n(T)$ the set of complete n-types over T.

Given a set of parameters A from a structure M, we can consider types in the language $L(A)$, which we call types over A. Here we stipulate that such types are consistent with $\mathrm{Th}(M, A)$, the theory of M in the language $L(A)$. Given a tuple b of length n from M or an elementary extension of M, the set of formulas in the language $L(A)$ true of b is a complete n-type over A, denoted $\mathrm{tp}(b/A)$. We write $S_n(A)$ for the set of complete n-types over A.

Proposition. *Given A a parameter set and b and c tuples, all in a model M,* $\mathrm{tp}(b/A) = \mathrm{tp}(c/A)$ *if and only if there is a $N \succeq M$ and an automorphism $f \in \mathrm{Aut}(N)$ such that $f : b \mapsto c$.*

By the compactness theorem, a set of formulas $\Sigma \subset L(A)$ is a type if and only if each finite subset of it is consistent. In this case, the complete theory $\mathrm{Th}(M, A)$ must include

$$\exists \overline{x} \, (\varphi_1(\overline{x}) \wedge \cdots \wedge \varphi_r(\overline{x}))$$

For each $\varphi_1, \ldots, \varphi_r \in \Sigma$. Therefore, types over A are *finitely realized* in M, any finite piece of them is satisfied by elements of M. We say that a type $p \in S_n(A)$ is *realized* in M if there is a tuple of elements b from M such that $\mathrm{tp}(b/A) = p$. For any $p \in S_n(A)$, there is a $N \succeq M$ such that p is realized in N.

Definition. For a cardinal κ, a structure M is said to be *κ-saturated* if for any $A \subset M$ with $|A| < \kappa$, all types in $S_n(A)$ are realized in M. A structure M is said to be *saturated* if it is a κ-saturated for $\kappa = |M|$.

In general, there is no guarantee that saturated models exist. For the theories ACF_p and DCF_0 we are considering, the fact that they are totally transcendental puts a bound on the size of $|S_n(A)|$ in terms of $|A|$ for any A, and this allows us to show that any model has a saturated elementary extension. The existence of fully saturated models, while not strictly necessary for the results in the paper, makes things more simple to state.

Models which are saturated or κ-saturated have two useful properties.

Definition. A model M of a complete theory T is *κ-universal* if any model $N \models T$ with $|N| < \kappa$ admits an elementary embedding into T.

Definition. A structure M is *κ-homogeneous* if given any partial elementary map $f : A \to M$ with $A \subset M$ of size $|A| < \kappa$, and any $b \in M$, we can extend f to a partial elementary map with domain $A \cup \{b\}$.

Proposition. *A κ-saturated model is κ-homogeneous and κ-universal.*

The significance of universality is that it allows us to study large classes of models as elementary substructures of a given one, and the significance of homogeneity is that it allows us to build automorphisms of the structure, one element at a time.

Proposition. *Given a saturated model M, $A \subset M$ a parameter set with $|A| < |M|$, and b, c tuples from M, the following are equivalent:*

(1) $\operatorname{tp}(b/A) = \operatorname{tp}(c/A)$

(2) *There is an automorphism f of M fixing A pointwise and sending b to c.*

Thus types over A correspond to orbits under the automorphism group $\operatorname{Aut}(M/A)$ of automorphisms of M fixing A. A consequence of this and compactness is the following.

Proposition (Basic Definability Theorem)**.** *If M is saturated and $X \subset M^n$ is a definable set which is stable under the action of* $\operatorname{Aut}(M/A)$*, X can be defined by a formula with parameters from A.*

These theorems allow us to use reasoning about automorphisms to prove things about definability, which has implications for all models of the theory.

Quantifier Elimination and Model Completeness. Quantifier elimination is a useful tool for understanding definable relations and types in particular theories by restricting the complexity of the formulas one needs to consider.

Definition. A theory T in a given language admits *quantifier elimination* if for any formula $\varphi(\overline{x})$, there is a quantifier free formula $\psi(\overline{x})$ such that

$$T \vdash (\forall \overline{x})(\varphi(\overline{x}) \leftrightarrow \psi(\overline{x}))$$

This property depends on the choice of language used to describe the desired class of structures. One can always add symbols to the language, and definitions of those symbols to the theory, such that the resulting theory eliminates quantifiers. In this paper we work with theories that admit quantifier elimination in their native languages, the languages of rings for ACF_p and differential rings for DCF_0.

One can in certain settings prove quantifier elimination using direct syntactic manipulation, or by considering automorphisms of the structure. In algebraic contexts, the following criterion is useful.

Proposition (Criterion for quantifier elimination)**.** *Let T be a theory in a language L. The following are equivalent:*

(1) *T admits quantifier elimination in the language L.*

(2) *For any A, B models of T where B is ω-saturated, sharing a common finitely generated L-substructure $C = \langle \overline{c} \rangle$ like so:*

$$\begin{tikzcd} A \arrow[dr, dash] & & B \arrow[dl, dash] \\ & C & \end{tikzcd}$$

if $\varphi(x, \overline{y})$ is quantifier free we have the implication

$$A \models (\exists x)\varphi(x, \overline{c}) \Longrightarrow B \models (\exists x)\varphi(x, \overline{c})$$

Proof. For the implication (1) $\implies$ (2), the existential formula $(\exists x)\varphi(x, \overline{c})$ is equivalent to a quantifier free formula $\psi(\overline{c})$ in both A and B. Because A and B contain $C = \langle \overline{c} \rangle$, they agree on the truth of ψ, so agree on the truth of $(\exists x)\varphi(x, \overline{c})$.

We now consider the implication (2) $\implies$ (1). To prove (1), it suffices to find quantifier free equivalents to formulas of the form $\exists x \varphi(x, \overline{y})$ where φ is quantifier free. This is because we can work outward from the subformulas of a given formula eliminating one quantifier at a time, and because $\forall$ is equivalent to $\neg\exists\neg$.

Let $\theta(\overline{y}) = (\exists x)\varphi(x, \overline{y})$. Let Γ be the set of all quantifier free consequences of θ,

$$\Gamma = \{\chi(\overline{y}) \mid \chi \text{ is quantifier free and } T, \theta(\overline{y}) \vdash \chi(\overline{y})\}$$

We will show that $T, \Gamma(\overline{y}) \vdash \theta(\overline{y})$. Let $B \models T$ and $\overline{c}$ a tuple from B be such that $B \models \Gamma(\overline{c})$, which is to say $B \models \chi(\overline{c})$ for all $\chi \in \Gamma$. To check that $B \models \theta(\overline{c})$, we may replace B with a ω-saturated elementary extension, and thereby assume that B itself is ω-saturated. Because we are assuming (2), to prove that $B \models \varphi(\overline{c})$, it suffices to find a model of T, $A \supset C = \langle \overline{c} \rangle$ such that $A \models \theta(\overline{c})$. For this, it suffices to show that

$$T \cup \Delta \cup \{\theta(\overline{c})\}$$

is consistent, where

$$\Delta = \{\delta(\overline{c}) \mid \delta \text{ is quantifier free and } B \models \delta(\overline{c})\}$$

This Δ is essentially the atomic diagram $D(C)$ of C, because each element of C is named by a term with parameters $\overline{c}$. Suppose to the contrary that $T \cup \Delta \cup \{\theta(\overline{c})\}$ is inconsistent. By compactness, for some finite $\Delta_0 \subset \Delta$,

$$T \cup \Delta_0 \cup \{\theta(\overline{c})\}$$

is inconsistent. Letting $\delta(\overline{c})$ be the conjunction of the elements of Δ_0, we then have that $T, \theta(\overline{c}) \vdash \neg\delta(\overline{c})$. There is nothing special about the constants $\overline{c}$, so $\neg\delta(\overline{y})$ is a consequence of $T, \theta(\overline{y})$. Therefore, $\neg\delta(\overline{y}) \in \Gamma$, which is a contradiction as $B \models \Gamma(\overline{c})$. We conclude that

$$T \cup \Delta \cup \{\theta(\overline{c})\}$$

is consistent, so we can find a model A. By (2), this implies that $B \models \theta(\overline{c})$. We conclude that $T, \Gamma(\overline{y}) \vdash \theta(\overline{y})$. By compactness, there is a finite $\Gamma_0 \subset \Gamma$ such that $T, \Gamma_0(\overline{y}) \vdash \theta(\overline{y})$. Let $\psi(\overline{y})$ be the conjunction of the elements of Γ_0. Then ψ is quantifier free, and

$$T \vdash \forall \overline{y}(\theta(\overline{y}) \leftrightarrow \psi(\overline{y}))$$

as required. This completes the proof of (1).

□

We will now introduce a property weaker than quantifier elimination, which has useful consequences.

Proposition (Theorem/Definition)**.** *The following are equivalent, for T a theory.*

(1) *Any formula $\varphi(\overline{x})$ is equivalent over T to an existential formula $(\exists \overline{y})\psi(\overline{x}, \overline{y})$, where ψ is quantifier free.*
(2) *Given models M, N of T with $M \subset N$, we have $M \preceq N$.*

We call a theory with such properties model complete.

It is clear from (1) that any theory admitting quantifier elimination is model complete. A useful consequence of model completeness is that whenever $M \subset N$ are models of T, A is a parameter set in N, and b is a tuple in M, $\text{tp}(b/A)$ is the same whether we evaluate it in M or in N. Examples of model complete theories include real closed fields and dense linear orders.